

Regular Expressions Practice (Regex)

Step-by-Step: How to Build a Regex Pattern

We can think of building regex like solving a *puzzle* — start simple, then get detailed.

Example 1: Finding Numbers in Text

What do all numbers have in common?

They are digits — 0 to 9 — and can appear one or more times.

- `\d` means “a digit”
- `+` means “one or more”

So the pattern becomes:

`\d+`

Meaning: Find one or more digits together.

Example 2: Finding Words Starting with Capital Letters

Ask: How does a capitalized word look?

It starts with a capital letter (A–Z), followed by lowercase letters (a–z).

- `[A-Z]` → any capital letter
- `[a-z]+` → one or more lowercase letters

Combine them:

`[A-Z][a-z]+`

This can help identify **names, cities, and organizations**.

Example 3: Finding Email Addresses

Ask: What do all emails have?

Letters or numbers before the `@` symbol, followed by a domain name.

Before `@` → `[A-Za-z0-9._]+`

`@` symbol → `@`

After `@` → `[A-Za-z]+\.`

Dot and domain type → `[A-Za-z]+`

Combine all:

`[A-Za-z0-9._]+@[A-Za-z]+\.[A-Za-z]+`

Meaning: Email address structure like *info@uok.edu.pk*.

Example 4: Finding Dates (like 12/10/2025)

Ask: *What do you notice about the structure?*

Two digits, a slash, two digits, a slash, then four digits.

- Two digits → `\d{2}`
- Slash → `/`
- Four digits → `\d{4}`

Combine them:

`\d{2}/\d{2}/\d{4}`

Meaning: Find dates in format *dd/mm/yyyy*.

Example 5: Finding Phone Numbers like +44-7890-123456

Ask: *How can we describe that shape?*

Optional plus sign, digits, and dashes.

Pattern:

`\+?\d{2,3}-\d{4}-\d{6}`

Meaning: Phone number with optional country code.

Example 6: Finding Money Values like \$250

Ask: *What's common?*

It starts with a dollar sign and is followed by digits.

Pattern:

`\$\d+`

Meaning: A dollar sign followed by one or more digits.

4. Practice

Read the text carefully, describe the pattern in words, and then form a regex for it.

Paragraph 1: Mr. Ahmed from London bought 3 items: Pen-001, Book-245, and Lamp-999. He paid £75.50 on 06/10/2025. Invoice was sent to ahmed.london@shopmail.com.

Paragraph 2: The conference on Data Science 2025 will be held in Islamabad, Pakistan. Registration fee is \$150 for students and \$300 for professionals. Visit www.datasummit.org or email info@datasummit.org for details.

Paragraph 3: Serial numbers include S-1001, S-1002, and S-1100. Each product is tested by Lab-A12 before packaging. Delivery date: 15/12/2025.

Paragraph 4: Student ID: STU-2301, Course: AI-102, Marks: 85%. Instructor: Dr. Smith, Department: Computer Science.

5. Reading and Explaining Regex (Reverse Practice)

Reading regex backwards helps in understanding.

For example: `\d{2}/\d{2}/\d{4}` → Two digits, slash, two digits, slash, four digits.

`[A-Z]{3}-\d{4}` → Three capital letters, dash, four digits.

`\$\d+(\.\d{2})?` → Dollar sign, digits, and optional decimal part.

6. How to Do

1. Always describe in words first, then translate into regex.
2. Escape special characters like \$, ., or ? when you want to search them literally.
3. Use quantifiers carefully — too broad can capture too much text, too narrow can miss matches.
4. Regex is not about memorization but pattern thinking.

7. Summary Table

Examples of regex patterns:

Numbers → `\d+`

Names → `[A-Z][a-z]+`

Dates → `\d{2}/\d{2}/\d{4}`

Product Codes → `[A-Z]{3}-\d{4}`

Prices → `[\$€]\d+(\.\d{2})?`

Percentages → `\d+%`

Emails → `[A-Za-z0-9._]+@[A-Za-z]+\.[A-Za-z]+`

Websites → `www\.[A-Za-z]+\.[A-Za-z]+`

Course Codes → `[A-Z]{2}-\d{3}`

Practice this on [regex101: build, test, and debug regex](#)